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ECA1004

- Microprocessors and Microcontrollers

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CO1 - Assessment Tool 2

(Constraint - Based Problem Solving)

50/50

Q1. ADD B

Given,

A = 5AH = 0101 1010

B = A6H = 1010 0110

1 0000 0000

Result : A = 00H

Flags :

S = 0

Z = 1

AC = 1

P = 1

CY = 1

Q2. Addressing Modes

Instruction	Addressing Mode	Operand Access
MVI A, 32H	Immediate	Data given in instruction
MOV A, B	Register	Operand in register B
LDA 2050H	Direct	Memory address directly given
MOV A, M	Indirect	Address obtained through HL

Trend: Immediate / register access uses readily available operands, while direct / indirect modes access memory.

Q3. SUB B

Given :

A = 65H = 0110 0101

B = 33H = 0011 0011

Using 2's complement:

33H → CCH → +1 → CDH

$$\begin{array}{r} 65H \\ +\ CDH \\ \hline 32H \end{array}$$

Result : A = 32H

Flags :

S = 0

Z = 0

AC = 0

P = 0

CY = 0

Q4. Speed vs Memory

Instruction	Memory Access	Speed
MOV A,B	No	Fastest
MOV A,M	Yes	Medium
LDA 2500H	Yes	Slowest

Conclusion : Register access is faster, while memory access increases execution time.

MOV A,B > MOV A,M > LDA 2500H

Q5. SBB B

Given :

A = 90H

B = 41H

CY = 1

Constraint : A - B - CY

90H - 41H - 1 = 4EH

Result : A = 4EH

Flags : S = 0

Z = 0

AC = 1

P = 0

CY = 0